Relations

9



Chapter Summary

9.1 Relations and Their Properties

9.2 n-ary Relations and Their Application

9.3 Representing Relations

9.4 Closures of Relations

9.5 Equivalence Relations

9.6 Partial Orderings



【Definition】A *binary relation* *R* from a set *A* to a set *B* is a subset of *A*×*B*.

Note:

- **A *binary relation R*** is a set.
- $R \subseteq A \times B$
- $R = \{(a,b) \mid a \in A, b \in B, aRb\}$

〖Example 1〗

(1) Let A and B be sets, ϕ, $A \times B$

(2) $A = \{2,3,4\}, B = \{2,3,4,5,6\}$ $R = \{(x,y) \mid x \in A, y \in B, x \mid y\}$

$R = \{(2,2),(2,4),(2,6),(3,3),(3,6),(4,4)\}$

(3) Let R be the set of real numbers,

$R_1 = \{(x,y) \mid x, y \in R, x^2 + y^2 \le 9\}$



1. Function As Relation

A function *f* from a set *A* to a set *B* is a relation from *A* to *B*.

〖Example 2〗

Suppose that $A=\{1,2,3,4\}, B=\{0,1\}$

***f*: *A*→*B*, *f*(1)=*f*(3)=1,*f*(2)=*f*(4)=0**

$R=\{(1,1),(3,1),(2,0),(4,0)\}$

Conversely, if *R* is a relation from *A* to *B*, is it a function?

〖Example 3〗

$R=\{(1,0),(1,1),(2,1),(3,0)\}$



2. Relations on a set

【Definition】A *relation on the set A* is a relation from *A* to *A*.

Note:

- $R \subseteq A \times A$

〖Example 4〗

(1) Let $A = \{1,2,3,4\}$, $R = \{(a,b) \mid a,b \in A, a \mid b\}$

(2) Suppose that *S* is a set. $R = \{(S_1, S_2) \mid S_1 \subseteq S_2, S_1, S_2 \in P(S)\}$

Question:

How many binary relations are there on a set *A* with *n* elements?

$$2^{n^2}$$



9.3 Representing Relations

The methods of representing relation:

- **list its all ordered pairs**
- **using a set build notation/specification by predicates**
- **2D table**
- **Connection matrix /zero-one matrix**
- **Directed graph/Digraph**



〖Example 5〗 $A = \{2,3,4\}, B = \{2,3,4,5,6\}$ $R = \{(x, y) \mid x \in A, y \in B, x \mid y\}$

$$R = \{(2,2),(2,4),(2,6),(3,3),(3,6),(4,4)\}$$

	2	3	4	5	6
2	×		×		×
3		×			×
4			×		



4. Connection Matrices

【Definition】Let *R* be a relation from

$$A=\{a_1,a_2,\cdots,a_m\},\quad to\quad B=\{b_1,b_2,\cdots,b_n\}$$

An $m\times n$ ***connection matrix*** $M_R=[m_{ij}]$ **for *R* is defined by**

$$m_{ij}=\begin{cases}1 & if\ (a_i,b_j)\in R,\\ 0 & if\ (a_i,b_j)\notin R.\end{cases}$$

For example, $A=\{2,3,4\}, B=\{2,3,4,5,6\}$ $R=\{(x,y)\mid x\in A, y\in B, x\mid y\}$

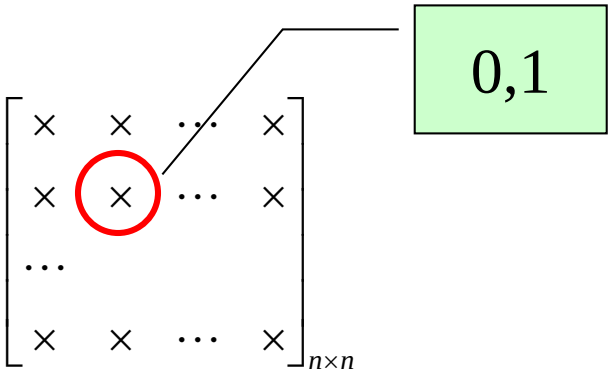
$$\begin{bmatrix}1&0&1&0&1\\0&1&0&0&1\\0&0&1&0&0\end{bmatrix}$$

$$\begin{array}{c} \\ 2\\3\\4\end{array}\begin{array}{c}\begin{array}{ccccc}2&3&4&5&6\end{array}\\\begin{bmatrix}1&0&1&0&1\\0&1&0&0&1\\0&0&1&0&0\end{bmatrix}\end{array}$$



Question:

How many binary relations are there on a set *A* with *n* elements?



By the product rule,

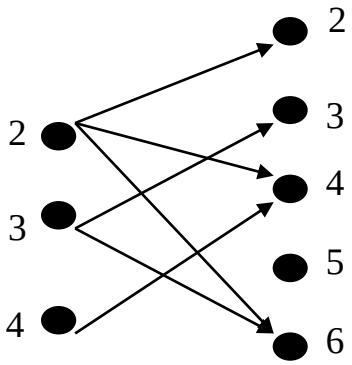
$$\overbrace{2 \times 2 \times \cdots \times 2}^{n^2} = 2^{n^2}$$

5. Directed graph/Digraph

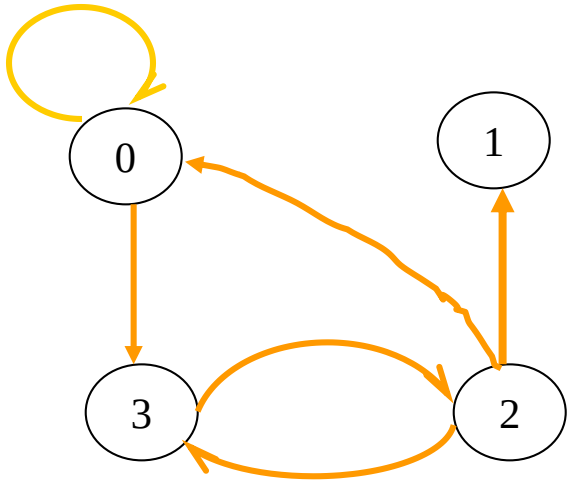
【Definition】A *directed graph* or a *digraph*, consists of a set *V* of *vertices* together with a set *E* of ordered pairs of elements of *V* called *edges(or arcs)*.

The *vertices a,b* is called the *initial* and *terminal* vertices of the edge *(a,b)*.

For example, $A = \{2,3,4\}, B = \{2,3,4,5,6\}$ $R = \{(x, y) \mid x \in A, y \in B, x \mid y\}$



〖Example 6〗 A = {0, 1, 2,3}



R = {(0,0), (0,3), (2,0),(2,1),(2,3),(3,2)}.

6. Special Properties of Binary Relations

- **Reflexive**
- **Irreflexive**
- **Symmetric**
- **Antisymmetric**
- **Transitive**



6. Special Properties of Binary Relations

- **Reflexive**
- Irreflexive
- Symmetric
- Antisymmetric
- Transitive



【Definition】 A relation R on a set A is *reflexive* if

$$(x,x)\in R,\ for\quad every\quad element\quad x\in A.$$

$$\forall x(x\in A\rightarrow (x,x)\in R)$$

Questions:

(1) What do we know about matrices representing reflexive relations?

All the elements on the main diagonal of $\mathbf{M}_R$ must be 1s.

$$M_r=\begin{bmatrix}1 & & & & \\ & 1 & & & \\ & & \cdot & & \\ & & & \cdot & \\ & & & & \cdot \\ & & & & & 1\end{bmatrix}$$

(2) What do we know about digraphs representing reflexive relations?

There is a loop at every vertex of the directed graph.



6. Special Properties of Binary Relations

- Reflexive
- **Irreflexive**
- Symmetric
- Antisymmetric
- Transitive



【Definition】**A relation *R* on a set *A* is *irreflexive* if**

$$\forall x(x \in A \rightarrow (x, x) \notin R)$$

Questions:

(1) The connection matrix of a irreflexive relations?

(2) Digraph?

(3) Can a relation on a set be neither reflexive nor irreflexive?

Yes.

$$\begin{bmatrix} 1 & \times & \cdots & \times \\ \times & 0 & \cdots & \times \\ \cdots & & & \\ \times & \times & \cdots & 0 \end{bmatrix}_{n \times n}$$



6. Special Properties of Binary Relations

- Reflexive
- Irreflexive
- **Symmetric**
- Antisymmetric
- Transitive

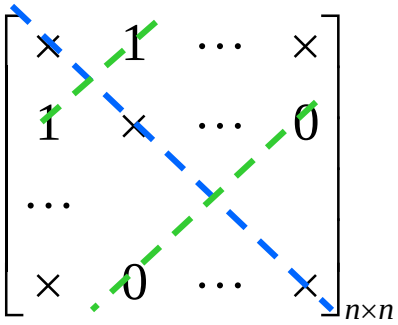


【Definition】**A relation *R* on a set *A* is *symmetric* if**

$$\forall x \forall y((x, y) \in R \rightarrow (y, x) \in R)$$

Questions:

(1) The connection matrix of a symmetric relations?



(2) Digraph?

If there is an arc (*x*, *y*) there must be an arc (*y*, *x*).

6. Special Properties of Binary Relations

- Reflexive
- Irreflexive
- Symmetric
- **Antisymmetric**
- Transitive



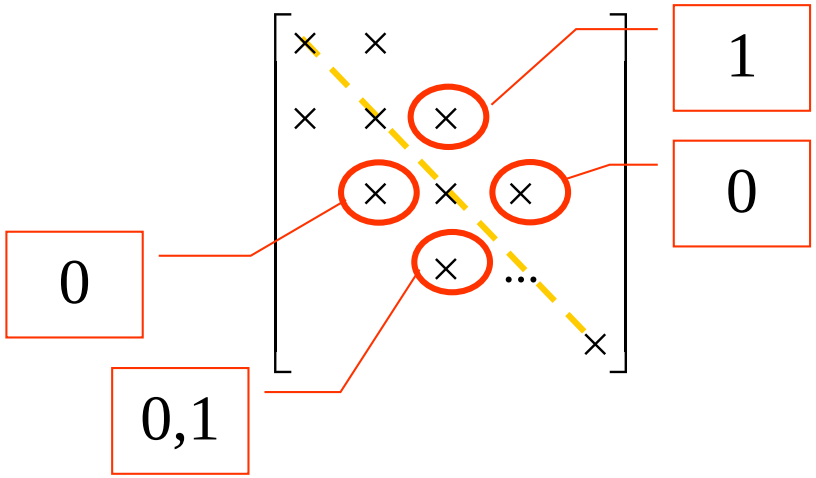
【Definition】 **A relation *R* on a set *A* is *antisymmetric* if**

$$\forall x \forall y((x, y) \in R \wedge (y, x) \in R \rightarrow x = y)$$

Note:

(1) $\forall x \forall y((x, y) \in R \wedge x \neq y \rightarrow (y, x) \notin R)$

(2)



(3) If there is an arc from x to y there cannot be one from y to x if $x \neq y$.

(4) The symmetric and antisymmetric relations are not opposites.

For example,

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$



6. Special Properties of Binary Relations

- Reflexive
- Irreflexive
- Symmetric
- Antisymmetric
- **Transitive**



【Definition】 **A relation *R* on a set *A* is *transitive* if whenever**

$$\forall x \forall y \forall z((x,y) \in R \wedge (y,z) \in R \rightarrow (x,z) \in R)$$

Note:

(1) $\overline{(r_{ij} \wedge r_{jk})} \vee r_{ik} = 1$

Why?

(2) If there is an arc from *x* to *y* and one from *y* to *z* then there must be one from *x* to *z*.

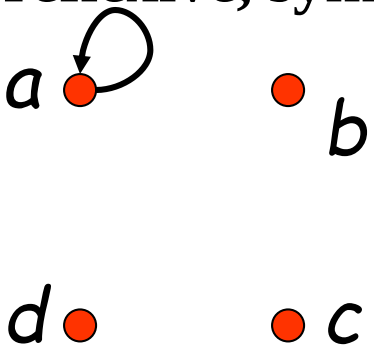


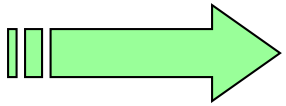
〖Example 7〗Determine whether the following relations are reflexive, irreflexive, symmetric, antisymmetric and/or transitive.

(1)
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

reflexive, symmetric, antisymmetric, transitive

(2)





$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

not reflexive, symmetric, antisymmetric, transitive

(3) $R_1 = \{(a,b) \mid a \mid b, a, b \in N\}$

reflexive, antisymmetric, transitive

(4) $R_2 = \{(a,b) \mid a + b = 2m, a, b, m \in N\}$

reflexive, symmetric, transitive

Question:

Symmetric, transitive $\Rightarrow$ reflexive ?

$$\left.\begin{array}{l} (a,b) \in R \\ R \ \ is \ \ symmetric \end{array}\right\} \Rightarrow \left.\begin{array}{l} (b,a) \in R \\ R \ \ is \ \ transitive \end{array}\right\} \Rightarrow (a,a) \in R$$

×



〖Example 8〗How many relations are there on a set with *n* elements that are

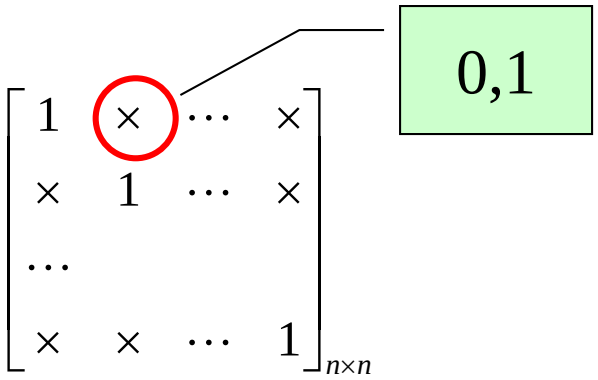
(1) reflexive ?

(2) symmetric ?

(3) antisymmetric ?

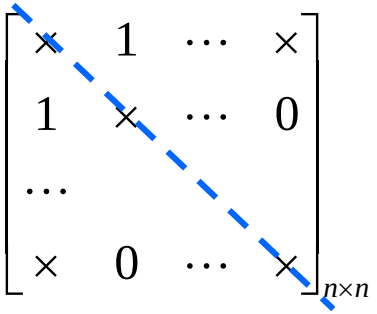
Solution:

(1)



$$2^{n^2-n}$$

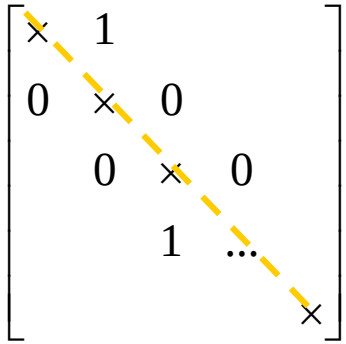
(2)



$$2^n \times 2^{\frac{n^2-n}{2}} = 2^{\frac{n(n+1)}{2}}$$

Solution:

(3)



$$2^n \times 3^{\frac{n^2-n}{2}}$$

Questions:

- **reflexive and symmetric?**
- **transitive?**

7. Combining Relations

Since relations form A to B are subsets of A×B, two relations form A to B can be combined in any way two sets can be combined.

- **Set operation**
- **Composition**
- **Inverse relation**



1) $\cup, \cap, \bar{}, -, \oplus$

〖Example 9〗 Let $A = \{1,2,3,4\}, Z$ **is the set of integers,**

$$R = \{(a,b) \mid a,b \in A, \frac{a-b}{2} \in Z\}, S = \{(a,b) \mid a,b \in A, \frac{a-b}{3} \in Z, a-b > 0\},$$

what are the relations $R \cup S, R \cap S, \bar{R}, R - S, S \oplus R$ **?**

Solution:

$$R = \{(1,1),(2,2),(3,3),(4,4),(1,3),(3,1),(2,4),(4,2)\} \qquad S = \{(4,1)\}$$

(1) Set operations

(2) Boolean operations/logical operations

The Boolean Sum $\vee : 0 \vee 0 = 0, 0 \vee 1 = 1, 1 \vee 0 = 1, 1 \vee 1 = 1$

The Boolean product $\wedge : 0 \wedge 0 = 0, 0 \wedge 1 = 0, 1 \wedge 0 = 0, 1 \wedge 1 = 1$

The complement $- : \bar{0} = 1, \bar{1} = 0$



The logical operations of matrices:

Let $A=\{a_1,a_2,\cdots,a_n\}, B=\{b_1,b_2,\cdots,b_m\}, M_{R_1}=[c_{ij}], M_{R_2}=[d_{ij}]$ **, the set operations of two relations are defined by**

$$M_{R_1\cup R_2}=[c_{ij}\vee d_{ij}]=M_{R_1}\vee M_{R_2}$$

$$M_{R_1\cap R_2}=[c_{ij}\wedge d_{ij}]=M_{R_1}\wedge M_{R_2}$$

$$M_{\overline{R}_1}=[\overline{c}_{ij}]$$

$$M_{R_1-R_2}=M_{R_1\cap\overline{R}_2}=[c_{ij}\wedge\overline{d}_{ij}]$$



2) Composition

$R = \{(a,b) \mid a \in A, b \in B, aRb\}, \quad S = \{(b,c) \mid b \in B, c \in C, bSc\}$

The ***composite of R and S***: $S \circ R$

$$S \circ R = \{(a,c) \mid a \in A \wedge c \in C \wedge \exists b(b \in B \wedge aRb \wedge bSc\}$$

Question:

How to compute *SoR*?**

(1) Using the definition directly

(2) Using the connection matrix



〖Example 10〗 $A=\{a,b\}, B=\{1,2,3,4\}, C=\{5,6,7\}$

$R=\{(a,1),(a,2),(b,3)\}, S=\{(2,6),(3,7),(4,5)\}$

$S\circ R=?\quad R\circ S=?$

Solution:

(1) Using the definition directly

$$S\circ R=\{(a,6),(b,7)\}$$

$$R\circ S=\phi$$

Note:

$$S\circ R\neq R\circ S$$



Solution:

(2) Using the connection matrix

$$\mathbf{M}_R = [r_{ij}]_{m\times n}, \mathbf{M}_S = [s_{jk}]_{n\times l}$$

$$\mathbf{M}_{S\circ R} = \mathbf{M}_R \cdot \mathbf{M}_S = [w_{ik}]_{m\times l}, \quad w_{ik} = \bigvee_{j=1}^{n}(r_{ij} \wedge s_{jk})$$

$$\because A = \{a, b\}, B = \{1,2,3,4\}, C = \{5,6,7\}$$

$$R = \{(a,1),(a,2),(b,3)\}, S = \{(2,6),(3,7),(4,5)\}$$

$$\therefore \mathbf{M}_R = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad \mathbf{M}_s = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{M}_{S\circ R} = \mathbf{M}_R \cdot \mathbf{M}_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore S \circ R = \{(a,6),(b,7)\}$$



【Definition】Let R be a relation on the set A. The powers $R^n, n = 1,2,3,\cdots$,are defined inductively by

$$R^1 = R, \quad and \quad R^{n+1} = R^n \circ R$$

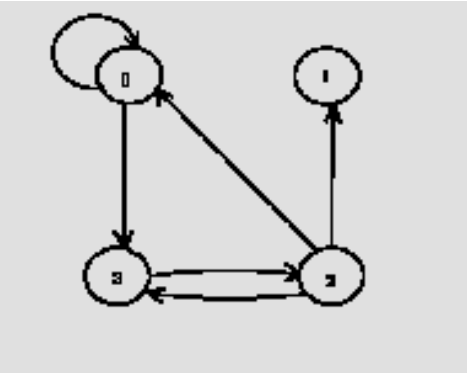
〖Example 11〗Let A={0,1,2,3}. R is the relation on the set A. R ={(0,0),(0,3),(2,3),(3,2),(2,1),(2,0)}. R^2=?

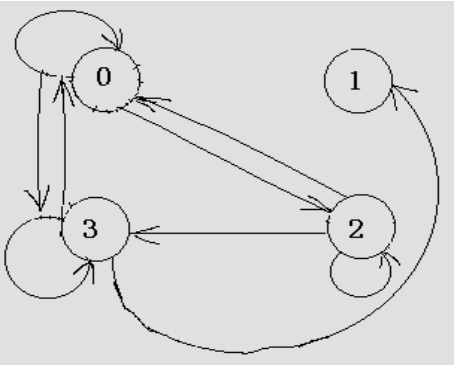
Solution:

(1) Using the definition

R^2={(0,0),(0,3),(0,2),(2,2),(3,3),(2,3),(2,0),(3,1),(3,0)}

(2) Using the digraph





(3) Using the matrix

$$\mathbf{M}_{R^2} = M_R \bullet M_R$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$



【 Theorem 】 The relation *R* on a set *A* is transitive if and only if $R^n \subseteq R, \quad for \quad n = 1,2,3,\cdots$

Proof:

(1) $R^n \subseteq R, \quad for \quad n = 1,2,3,\cdots \Rightarrow$ **R is transitive**

$(a,b) \in R, \ (b,c) \in R \Rightarrow (a,c) \in R$

$\Downarrow$

$(a,c) \in R^2 \subseteq R$

(2) *R* is transitive $\Rightarrow \quad R^n \subseteq R, \quad for \quad n = 1,2,3,\cdots$

- **Inductive base** $n = 1, \quad R \subseteq R$
- **Inductive step** $R^n \subseteq R \quad \Rightarrow \quad R^{n+1} \subseteq R$

$(a,b) \in R^{n+1}$

$R^{n+1} = R^n \circ R$

$\Rightarrow$

$(a,x) \in R, (x,b) \in R^n \subseteq R$

R is transitive

$\Rightarrow (a,b) \in R$



〖Example 12〗 Let R be a symmetric relation on the set A. Show that R^n is symmetric for all positive integers n.

Proof:

- **Inductive base**

  n=1, R be a symmetric

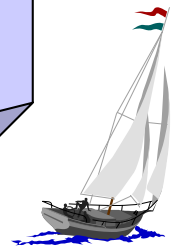
- **Inductive step**

  R^n is symmetric $\Rightarrow$ R^{n+1} is symmetric

$$(a,c) \in R^{n+1} \Rightarrow (c,a) \in R^{n+1}$$

$$\left.\begin{array}{l}(a,b) \in R, (b,c) \in R^n \\ R, R^n \ \ are \ \ symmetric\end{array}\right\} \Rightarrow (b,a) \in R, (c,b) \in R^n$$

$$\Rightarrow (c,a) \in R \circ R^n = R^{n+1}$$



3) Inverse relation

$$R = \{(a,b) \mid a \in A, b \in B, aRb\}$$

The *inverse relation* form *B* to *A*: $R^{-1}(R^c)$

$$\{(b,a) \mid (a,b) \in R, a \in A, b \in B\}$$

Question:

How to get R^{-1} **?**

(1) Using the definition directly

For example, $R = \{(a,b) \mid a \mid b, a, b \in Z^+\}$

$R^{-1} = \{(a,b) \mid b \mid a, a, b \in Z^+\}$

(2) Reverse all the arcs in the digraph representation of *R*

(3) Take the transpose $M_R{}^T$ of the connection matrix M_R of *R*.



4) The properties of relation operations

Suppose that *R*, *S* are the relations from *A* to *B*, *T* is the relation from *B* to *C*, *P* is the relation from *C* to *D*, then

(1) $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$

Proof:

$$\forall (x, y) \in (R \cup S)^{-1}$$

$$\Leftrightarrow (y, x) \in R \cup S$$

$$\Leftrightarrow (y, x) \in R \quad or \quad (y, x) \in S$$

$$\Leftrightarrow (x, y) \in R^{-1} \quad or \quad (x, y) \in S^{-1}$$

$$\Leftrightarrow (x, y) \in R^{-1} \cup S^{-1}$$



4) The properties of relation operations

Suppose that *R*, *S* are the relations from *A* to *B*, *T* is the relation from *B* to *C*, *P* is the relation from *C* to *D*, then

(1) $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$

(2) $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$

(3) $(\overline{R})^{-1} = \overline{R^{-1}}$

(4) $(R - S)^{-1} = R^{-1} - S^{-1}$

(5) $(A \times B)^{-1} = B \times A$

Proof:

$$\forall (x, y) \in (A \times B)^{-1}$$

$$\Leftrightarrow \quad (y, x) \in A \times B$$

$$\Leftrightarrow \quad (x, y) \in B \times A$$



4) The properties of relation operations

Suppose that *R*, *S* are the relations from *A* to *B*, *T* is the relation from *B* to *C*, *P* is the relation from *C* to *D*, then

(1) $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$

(2) $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$

(3) $(\overline{R})^{-1} = \overline{R^{-1}}$

(4) $(R - S)^{-1} = R^{-1} - S^{-1}$

(5) $(A \times B)^{-1} = B \times A$

(6) $\overline{R} = A \times B - R$

(7) $(S \circ T)^{-1} = T^{-1} \circ S^{-1}$

(8) $(R \circ T) \circ P = R \circ (T \circ P)$

(9) $(R \cup S) \circ T = R \circ T \cup S \circ T$



Homework:

P. 608: 6, 25, 30, 47, 50, 60



Chapter 9 Relations

9.1 Relations and Their Properties

9.2 n-ary Relations and Their Application

9.3 Representing Relations

9.4 Closures of Relations

9.5 Equivalence Relations

9.6 Partial Orderings



9.2 n-ary Relations and Their Applications

【Definition】 Let $A_1, A_2, \cdots, A_n$ be sets. An *n*-ary Relation on these sets is a subset of $A_1 \times A_2 \times \cdots \times A_n$

Example: Let *R* be the relation on Z × Z × Z+ consisting of triples *(a, b,m)*, where *a*, *b*, and *m* are integers with $m \geq 1$ and $a \equiv b \pmod{m}$. Then *(*8, 2, 3*)*, *(*−1, 9, 5*)*, and *(*14, 0, 7*)* all belong to *R*, but *(*7, 2, 3*)*, *(*−2,−8, 5*)*, and *(*11, 0, 6*)* do not belong to *R* because $8 \equiv 2 \pmod{3}$, $-1 \equiv 9 \pmod{5}$, and $14 \equiv 0 \pmod{7}$, but $7 \neq 2 \pmod{3}$, $-2 \neq -8 \pmod{5}$, and $11 \neq 0 \pmod{6}$.



9.2 n-ary Relations and Their Applications

【Definition】The projection $P_{i1i2,\cdots,im}$ **maps the *n*-tuple** $(a_1, a_2, \cdots, a_n)$ **to the *m*-tuple** $(a_{i1}, a_{i2}, \cdots, a_{im})$ **where** $m \leq n$

【Definition】Let *R* be a relation of degree *m* and *S* a relation of degree *n*. The *join Jp(R, S)*, where $p \leq m$ and $p \leq n$, is a relation of degree $m + n - p$ that consists of all $(m + n - p)$-tuples $(a1, a2, \ldots, am-p, c1, c2, \ldots, cp, b1, b2, \ldots, bn-p)$, where the *m*-tuple $(a1, a2, \ldots, am-p, c1, c2, \ldots, cp)$ belongs to *R* and the *n*-tuple $(c1, c2, \ldots, cp, b1, b2, \ldots, bn-p)$ belongs to *S*.



9.2 n-ary Relations and Their Applications

EXAMPLE 8 What results when the projection $P_{1,3}$ is applied to the 4-tuples (2, 3, 0, 4), (Jane Doe, 234111001, Geography, 3.14), and (a_1, a_2, a_3, a_4)?

Solution: The projection $P_{1,3}$ sends these 4-tuples to (2, 0), (Jane Doe, Geography), and (a_1, a_3), respectively. ◀

Example 9 illustrates how new relations are produced using projections.

EXAMPLE 9 What relation results when the projection $P_{1,4}$ is applied to the relation in Table 1?

Solution: When the projection $P_{1,4}$ is used, the second and third columns of the table are deleted, and pairs representing student names and grade point averages are obtained. Table 2 displays the results of this projection. ◀

TABLE 1 Students.

Student_name	*ID_number*	*Major*	*GPA*
Ackermann	231455	Computer Science	3.88
Adams	888323	Physics	3.45
Chou	102147	Computer Science	3.49
Goodfriend	453876	Mathematics	3.45
Rao	678543	Mathematics	3.90
Stevens	786576	Psychology	2.99

TABLE 2 GPAs.

Student_name	*GPA*
Ackermann	3.88
Adams	3.45
Chou	3.49
Goodfriend	3.45
Rao	3.90
Stevens	2.99



9.2 n-ary Relations and Their Applications

TABLE 5 Teaching_assignments.

Professor	*Department*	*Course_number*
Cruz	Zoology	335
Cruz	Zoology	412
Farber	Psychology	501
Farber	Psychology	617
Grammer	Physics	544
Grammer	Physics	551
Rosen	Computer Science	518
Rosen	Mathematics	575

TABLE 6 Class_schedule.

Department	*Course_number*	*Room*	*Time*
Computer Science	518	N521	2:00 P.M.
Mathematics	575	N502	3:00 P.M.
Mathematics	611	N521	4:00 P.M.
Physics	544	B505	4:00 P.M.
Psychology	501	A100	3:00 P.M.
Psychology	617	A110	11:00 A.M.
Zoology	335	A100	9:00 A.M.
Zoology	412	A100	8:00 A.M.

What relation results when the join operator $J2$ is used to combine the relation displayed in Tables 5 and 6?

TABLE 7 Teaching_schedule.

Professor	*Department*	*Course_number*	*Room*	*Time*
Cruz	Zoology	335	A100	9:00 A.M.
Cruz	Zoology	412	A100	8:00 A.M.
Farber	Psychology	501	A100	3:00 P.M.
Farber	Psychology	617	A110	11:00 A.M.
Grammer	Physics	544	B505	4:00 P.M.
Rosen	Computer Science	518	N521	2:00 P.M.
Rosen	Mathematics	575	N502	3:00 P.M.

Chapter 9 Relations

9.1 Relations and Their Properties

9.2 n-ary Relations and Their Application

9.3 Representing Relations

9.4 Closures of Relations

9.5 Equivalence Relations

9.6 Partial Orderings



9.4 Closures of Relations

- **Reflexive Closure**
- **Symmetric Closure**
- **Transitive Closure**

【Definition】*The closure of a relation R* with respect to property *P* is the relation *S* with property *P* containing *R* such that *S* is a subset of every relation with property *P* containing *R*.

The smallest relation with property *P* containing *R*



1. Reflexive Closure

【 Theorem 】 Let *R* be a relation on *A*. The *reflexive closure* of *R*, denoted by *r*(*R*), is $R \cup I_A$

The *diagonal relation* on *A*

$$I_A = \{(x, x) \mid x \in A\}$$

Proof:

① **containing *R***

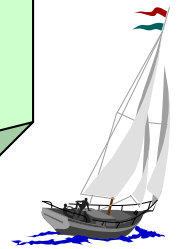
② **is a reflexive relation**

$$\forall x \in A, (x, x) \in I_A \subseteq R \cup I_A$$

③ **is the smallest reflexive relation which contains *R***

Suppose that *R*′ is a reflexive relation containing *R*, then

$$R \subseteq R', I_A \subseteq R' \Rightarrow r(R) = R \cup I_A \subseteq R'$$



【 Corollary 】 $R = R \cup I_A \Leftrightarrow$ ***R* is a reflexive relation .**

Proof:

① **The reflexive closure is a reflexive relation**

② **Since *R* is a reflexive relation,**

$I_A \subseteq R$

$\therefore R \cup I_A = R$



Question:

Given *R*, how to obtain its reflexive closure?

$$r(R) = R \cup I_A$$

- **Add to *R* all ordered pairs of the form (*a*,*a*) with $a \in A$, not already in *R***
- **Add loops to all vertices on the digraph representation of *R*.**
- **Put 1's on the diagonal of the connection matrix of *R*.**



〖Example 1〗 $R=\{(a,b)\mid a<b,a,b\in Z\}$ **, What is *r*(*R*)?**

Solution:

$$r(R)=R\bigcup I_A$$

$$=\{(a,b)\mid a<b,a,b\in Z\}\bigcup\{(a,a)\mid a\in Z\}$$

$$=\{(a,b)\mid a\le b,a,b\in Z\}$$



2. Symmetric Closure

【 Theorem 】 **Let *R* be a relation on *A*. The *symmetric closure* of *R*, denoted by *s*(*R*), *is*** $R \cup R^{-1}$

Proof:

① **containing *R***

② **is a symmetric relation**

$$(a,b) \in R \cup R^{-1} \Rightarrow \begin{cases} (a,b) \in R & \Rightarrow (b,a) \in R^{-1} \\ (a,b) \in R^{-1} & \Rightarrow (b,a) \in R \end{cases}$$

$$\Rightarrow (b,a) \in R \cup R^{-1}$$



③ **is the smallest symmetric relation which containing *R***

Suppose that *R′* is a symmetric relation containing *R*, then

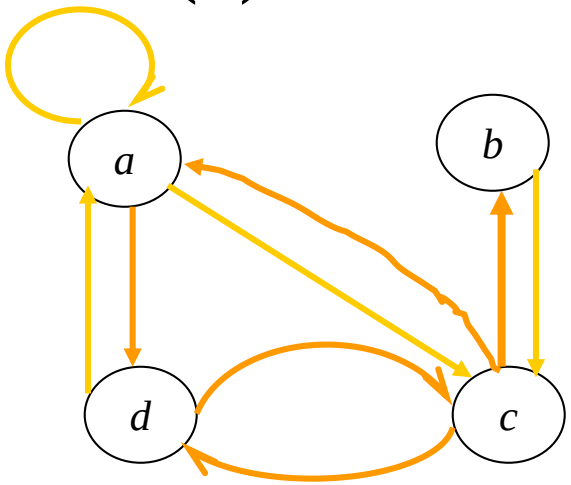
If $(a,b) \in R \cup R^{-1}$

$$\Rightarrow \begin{cases} \left.\begin{array}{l}(a,b) \in R \\ R \subseteq R'\end{array}\right\} \Rightarrow (a,b) \in R' \\ (a,b) \in R^{-1} \Rightarrow (b,a) \in R \Rightarrow (b,a) \in R' \\ \qquad\qquad\qquad\qquad \left.\begin{array}{l} \\ R' \text{ is a symmetric relation}\end{array}\right\} \Rightarrow (a,b) \in R' \end{cases}$$

$$\Rightarrow R \cup R^{-1} \subseteq R'$$



〖Example 2〗 What is $s(R)$ of the following relation?



Note:

- Add an edge from x to y whenever this edge is not already in directed graph but the edge from y to x is.
- Add all ordered pairs of the form (b,a) where (a,b) is in the relation, that are not already in R.
- $M_{s(R)} = M_R \vee M_R^T$

【Corollary】$R = R \bigcup R^{-1} \Leftrightarrow$ *R* is a symmetric relation .

Proof:

① **The symmetric closure is a symmetric relation**

② **Since *R* is a symmetric relation, it follows that**

$$(a,b) \in R \Rightarrow (b,a) \in R$$

$$\Rightarrow R^{-1} \subseteq R$$

$$\Rightarrow R^{-1} \bigcup R = R$$



3. Transitive Closure

Terminology:

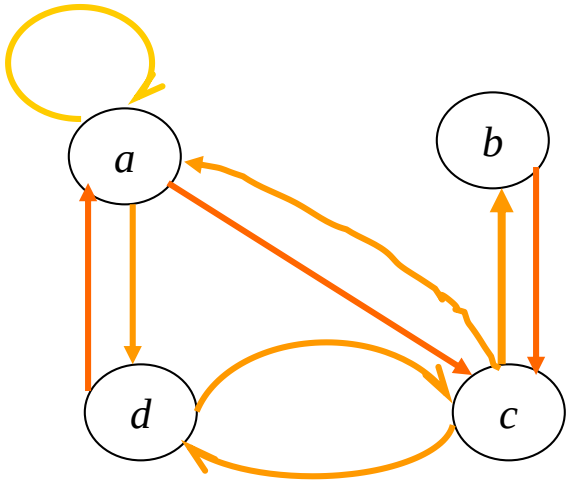
***A path of length n in a digraph G* :**

A sequence of edges $(x_0, x_1),(x_1, x_2),\ldots,(x_{n-1}, x_n)$

Notation: $x_0, x_1, x_2, \ldots, x_{n-1}, x_n$.

***Cycle or circuit* :** **If $x_0 = x_n$**

For example,



The term path also applies to relation.

Why?

Relation ↔ digraph

How?

There is a path of length *n* from *a* to *b* in *R*

$\exists a, x_1, x_2, \ldots, x_{n-1}, b$ **such that** $(a, x_1) \in R, (x_1, x_2) \in R, \ldots, (x_{n-1}, b) \in R$



【 Theorem 】 Let *R* be a relation on *A*. There is a path of length *n* from *a* to *b* if and only if $(a,b) \in R^n$

Proof:

① **Inductive basis**

An edge from *a* to *b* is a path of length 1 which is in $R^1 = R$. Hence the assertion is true for $n = 1$.

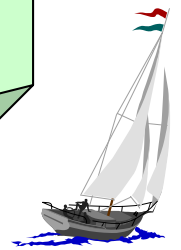
② **Inductive step**

There is a path of length $n+1$ from *a* to *b* if and only if there is an *x* in *A* such that there is a path of length 1 from *a* to *x* and a path of length *n* from *x* to *b*.

From the Induction Hypothesis,

$$(a,x) \in R \qquad (x,b) \in R^n$$

$$(a,b) \in R^n \circ R = R^{n+1}$$



Recall:

The transitive closure of a relation R, $t(R)$, is the smallest transitive relation containing R.

How can we construct $t(R)$?

Can $t(R)$ be produced by adding all the pairs of the form (a, c) where (a, b) and (b, c) are already in the relation?

〖Example 3〗 Suppose that $R = \{(1,2),(2,2),(3,1)\}$ be a relation on the set $A = \{1,2,3\}$

$t(R)$=?
How to computer?

$$R' = \{(1,2),(2,3),(3,1),(1,3),(2,1),(3,2)\}$$

$$R' = t(R)\,?$$



【Definition】 ***The connectivity relation*** **denoted by *R**, is the set of ordered pairs (*a*, *b*) such that there is a path (in *R*) from *a* to *b*:**

$$R^* = \bigcup_{n=1}^{\infty} R^n$$

〖Example 3〗 Let *A* = the set of people, *R* = {(*x*, *y*) | person *x* is a parent of person *y*}. *R = ?**



【 Theorem 】 ***t*(*R*) = *R**.**

Proof:

① **containing *R***

② **is a transitive relation**

Suppose (*a*, *c*) and (*c*, *b*) are in *R. Show that (*a*, *b*) is in *R**.**

By the definition of *R,**

$$(a,c),(c,b) \in R^*$$

$$\Rightarrow \exists i, j \quad (a,c) \in R^i, (c,b) \in R^j$$

$$\Rightarrow (a,b) \in R^{i+j} \subseteq R^*$$



③ **is the smallest transitive relation which contains R**

Now suppose that S is any transitive relation which contains R. We must show S contains R^* to show R^* is the smallest relation.

Since S is transitive, S^n also is transitive and $S^n \subseteq S$.

Furthermore, since $S^* = \bigcup_{k=1}^{\infty} S^k$

and $S^n \subseteq S$, it follows that $S^* \subseteq S$

If $R \subseteq S$, then $R^* \subseteq S^*$, because any path in R is also a path in S.

In fact, we need only consider paths of length n or less.



【 Theorem 】 If $|A| = n$, then any path of length $> n$ must contain a cycle.

Proof:

If we write down a list of more than *n* vertices representing a path in *R*, some vertex must appear at least twice in the list (by the Pigeon Hole Principle).

$$a = x_0,\ x_1,\ x_2, \cdots,\ x_{i-1}, \boxed{x_i},\ x_{i+1}, \cdots,\ x_{j-1}, \boxed{x_j},\ x_{j+1},\ x_{j+2}, \cdots, x_m = b \qquad m > n$$



【Theorem】 **If** $|A| = n$ **, R is a relation on A, then**

$$\exists k, k \le n, R^* = R \cup R^2 \cup \cdots \cup R^k$$

【Corollary】 **If** $|A| = n$ **, then** $t(R) = R^* = R \cup R^2 \cup \cdots \cup R^n$

【Corollary】 **Let** $\mathbf{M}_R$ **be the zero-one matrix of the relation R on a set with n elements. The zero-one matrix of the transitive closure is**

$$\mathbf{M}_{t(R)} = \mathbf{M}_R \vee \mathbf{M}_R^{[2]} \vee \cdots \vee \mathbf{M}_R^{[n]}$$



A Procedure for computing $t(R)$

$$\mathbf{M}_{t(R)} = \mathbf{M}_R \vee \mathbf{M}_R^{[2]} \vee \cdots \vee \mathbf{M}_R^{[n]}$$

$A = \mathbf{M}_R$;

$B = A$;

for (*i*=2; *i*<=n; *i*++)

{

$A = A \cdot \mathbf{M}_R$;

$B := B \vee A$;

}

The complexity of algorithm:

$$n^2(2n-1)(n-1) + (n-1)n^2 = 2n^3(n-1) = O(n^4)$$



Warshall's Algorithm

The interior vertices of a path:

$$x_0,\quad x_1,\quad x_2,\quad \cdots\quad ,\quad x_{n-1},\quad x_n$$

Warshall's algorithm is based on the construction of a sequence of zero-one matrices, such as

$$W_0,\quad W_1,\quad W_2,\quad \cdots\quad ,W_n$$

$$W_0 = M_R$$

$$W_k = [w_{ij}^{(k)}]$$

$$w_{ij}^{(k)} = \begin{cases} 1 & \text{If there is a path from } V_i \text{ to } V_j \text{ such that all the interior vertices of this path} \\ & \text{are in the set } \{V_1, V_2, \cdots, V_k\} \\ 0 & \text{otherwise} \end{cases}$$

$$W_n = M_{t(R)}$$



EXAMPLE 8 Let R be the relation with directed graph shown in Figure 3. Let a, b, c, d be a listing of the elements of the set. Find the matrices $\mathbf{W}_0$, $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{W}_3$, and $\mathbf{W}_4$. The matrix $\mathbf{W}_4$ is the transitive closure of R.

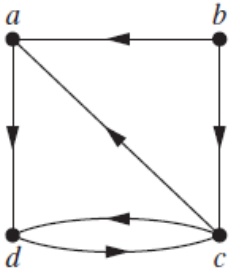


FIGURE 3 **The Directed Graph of the Relation R.**

Solution: Let $v_1 = a$, $v_2 = b$, $v_3 = c$, and $v_4 = d$. $\mathbf{W}_0$ is the matrix of the relation. Hence,

$$\mathbf{W}_0 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

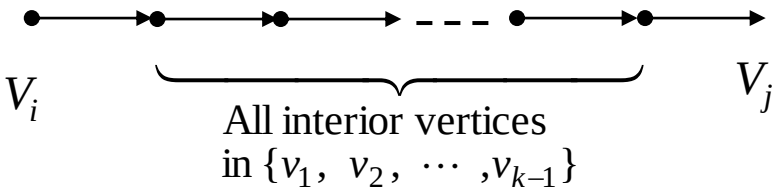
$\mathbf{W}_1$ has 1 as its (i, j)th entry if there is a path from v_i to v_j that has only $v_1 = a$ as an interior vertex. Note that all paths of length one can still be used because they have no interior vertices. Also, there is now an allowable path from b to d, namely, b, a, d. Hence,

$$\mathbf{W}_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

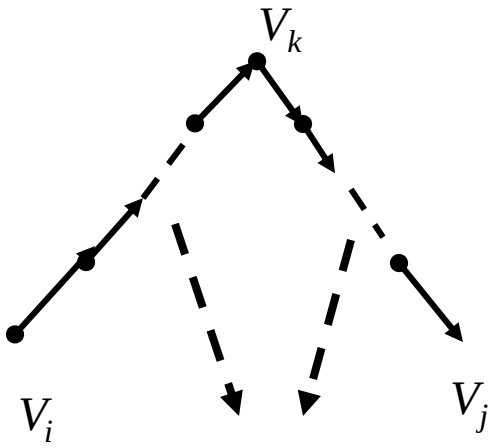
$\mathbf{W}_2$ has 1 as its (i, j)th entry if there is a path from v_i to v_j that has only $v_1 = a$ and/or $v_2 = b$ as its interior vertices, if any. Because there are no edges that have b as a terminal vertex, no new paths are obtained when we permit b to be an interior vertex. Hence, $\mathbf{W}_2 = \mathbf{W}_1$.

$$w_{ij}^{(k)} = w_{ij}^{(k-1)} \vee \left(w_{ik}^{(k-1)} \wedge w_{kj}^{(k-1)} \right)$$

Case 1



Case 2



Warshall's Algorithm

$W = [w_{ij}]_{n\times n}$;
for (*k*=1; *k* <=*n*; *k* ++)
{
for (*i*=1; *i* <=*n*; *i* ++)
{
for (*j*=1; *j* <=*n*; *j* ++)

$$w_{ij} = w_{ij} \vee (w_{ik} \wedge w_{kj});$$

}
}

The complexity of algorith

if ($w_{ik} = 1$ **)**
{
for (*j*=1; *j* <=*n*; *j* ++)

$$w_{ij} = w_{ij} \vee w_{kj};$$

}



〖Example〗 Let

$$A=\{1,2,3,4,5\}, R=\{(1,1),(1,2),(2,4),(3,5),(4,2)\}, t(R)=?$$

Solution:

$$M=\begin{bmatrix}1&1&0&0&0\\0&0&0&1&0\\0&0&0&0&1\\0&1&0&0&0\\0&0&0&0&0\end{bmatrix}$$

$k=1$ →
$$\begin{bmatrix}1&1&0&0&0\\0&0&0&1&0\\0&0&0&0&1\\0&1&0&0&0\\0&0&0&0&0\end{bmatrix}$$

$k=2$ →
$$\begin{bmatrix}1&1&0&1&0\\0&0&0&1&0\\0&0&0&0&1\\0&1&0&1&0\\0&0&0&0&0\end{bmatrix}$$

$k=3$ →
$$\begin{bmatrix}1&1&0&1&0\\0&0&0&1&0\\0&0&0&0&1\\0&1&0&1&0\\0&0&0&0&0\end{bmatrix}$$

$k=4$ →
$$\begin{bmatrix}1&1&0&1&0\\0&1&0&1&0\\0&0&0&0&1\\0&1&0&1&0\\0&0&0&0&0\end{bmatrix}$$

$k=5$ →
$$\begin{bmatrix}1&1&0&1&0\\0&1&0&1&0\\0&0&0&0&1\\0&1&0&1&0\\0&0&0&0&0\end{bmatrix}$$



Homework:

P. 637： 3, 20, 23, 26(d), 28(d), 29



Chapter 9 Relations

9.1 Relations and Their Properties

9.2 n-ary Relations and Their Application

9.3 Representing Relations

9.4 Closures of Relations

9.5 Equivalence Relations

9.6 Partial Orderings



9.5 Equivalence Relations

1. Equivalence relations and equivalence classes

【Definition】**A relation *R* on a set *A* is an *equivalence relation* if *R* is**

- **reflexive**
- **symmetric**
- **transitive**

For example,

(1) $\{(a,b) \mid a+b=2m, a,b,m \in N\}$

(2) The similarity relation between two triangles

(3) The equivalent relation between two formulas in proposition logic



Terminologies:

a and b are equivalent

R* is an equivalence relation, and (*a* , *b*)∈ *R

the equivalence class of x

The set of all elements that are related to an element *x* of *A*

Notation: $[x]_R$ $[x]$

a representative of the equivalence class $[x]_R$: $b \in [x]_R$



〖Example 1〗 Congruence Modulo 3

$$R = \{(a,b) \mid a \equiv b (\mathrm{mod}\ 3), a, b \in Z\}$$

Show that *R* is an equivalence relation. And find its equivalence class.

Solution:

$$a \equiv b (\mathrm{mod}\ 3) \text{ if and only if } 3 \mid (a-b)$$

① **reflexive**

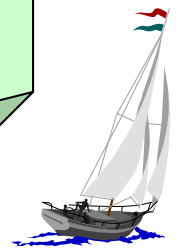
***R* is reflexive, since** $3 \mid (a-a)$

② **symmetric**

$$(a,b) \in R \Rightarrow a \equiv b (\mathrm{mod}\ 3) \Rightarrow 3 \mid (a-b) \Rightarrow a-b = 3k, k \in Z$$
$$\Rightarrow b-a = -3k \Rightarrow b \equiv a (\mathrm{mod}\ 3) \Rightarrow (b,a) \in R$$

③ **transitive**

$$3 \mid (a-b), 3 \mid (b-c)$$
$$a-c = (a-b)+(b-c) \Rightarrow 3 \mid (a-c)$$



[0]={…, -6, -3, 0, 3, 6, …}={ $3k \mid k \in Z$ }

[1]={…, -5, -2, 1, 4, 7, …}={ $3k+1 \mid k \in Z$ }

[2]={…, -4, -1, 2, 5, 8, …}={ $3k+2 \mid k \in Z$ }

Congruence Modulo *m* :

$$R = \{(a,b) \mid a \equiv b (\mathrm{mod}\ m), a, b \in Z\}$$

Congruence class Modulo *m* :

$$[0]_m, [1]_m \cdots, [m-1]_m$$

$$[a]_m = \{\cdots, a-2m, a-m, a, a+m, a+2m, \cdots\}$$



〖Example 2〗 Suppose that *A* is a nonempty set, and *f* is a function that has *A* as its domain. Let *R* be the relation on *A* consisting of all ordered pairs (*x*,*y*) where *f*(*x*)=*f*(*y*).

(1) Show that *R* is an equivalence relation.

(2) What are the equivalence class of *R*.

Solution:

(1) $R = \{(x, y) \mid x, y \in A \wedge f(x) = f(y)\}$

① **reflexive**

② **symmetric**

③ **transitive**

(2) $[x] = \{y \mid y \in A \wedge f(x) = f(y)\}$

$\{f^{-1}(b) \mid b \in f(A)\}$



Example: Suppose that R is the relation on the set of strings of English letters such that aRb if and only if $l(a) = l(b)$, where $l(x)$ is the length of the string x. Is R an equivalence relation?

Solution: Show that all of the properties of an equivalence relation hold.

***Reflexivity*: Because $l(a) = l(a)$, it follows that aRa for all strings a.**

***Symmetry*: Suppose that aRb. Since $l(a) = l(b)$, $l(b) = l(a)$ also holds and bRa.**

***Transitivity*: Suppose that aRb and bRc. Since $l(a) = l(b)$,and $l(b) = l(c)$, $l(a) = l(c)$ also holds and aRc.**



2. Equivalence relations and partitions

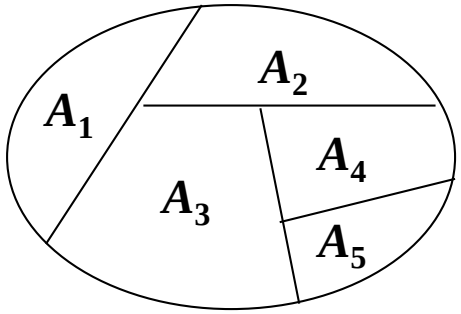
【Definition】 **Let** $\{A_1, A_2, \cdots\}$ **be a collection of subsets of** A**. Then the collection forms a *partition* of** A **if and only if**

- $A_i \neq \phi$ for $i \in Z$
- $A_i \cap A_j = \phi$, when $i \neq j$
- $\forall a \in A, \exists i$ such that $a \in A_i (i = 1,2,\cdots)$

$\bigcup_{i \in Z} A_i = A$

Notation:

$$pr(A) = \{A_1, A_2, \cdots\}$$



【Theorem 1】 Let R be an equivalence relation on a set A. The following statements are equivalent:

(1) aRb

(2) $[a]=[b]$

(3) $[a]\cap[b]\neq\phi$

Proof:

- **Show that (1) implies (2)**

$$[a]=[b]\Leftrightarrow([a]\subseteq[b])\wedge([b]\subseteq[a])$$

$$\left.\begin{array}{l} x\in[a]\Rightarrow(a,x)\in R \\ aRb\Rightarrow(a,b)\in R\Rightarrow(b,a)\in R \end{array}\right\}\Rightarrow(b,x)\in R\Rightarrow x\in[b]\Rightarrow[a]\subseteq[b]$$



(1) aRb
(2) $[a]=[b]$
(3) $[a]\cap[b]\neq\phi$

- **Show that (2) implies (3)**

$$\left.\begin{array}{l}[a]=[b]\\ \textbf{\textit{R} is reflexive} \Rightarrow \textbf{[}a\textbf{] is nonempty}\end{array}\right\} \Rightarrow [a]\cap[b]\neq\phi$$

- **Show that (3) implies (1)**

$$[a]\cap[b]\neq\phi \quad \Rightarrow \exists x \in [a]\cap[b]$$

$$\Rightarrow (a,x)\in R, (b,x)\in R$$

$$\Rightarrow (a,b)\in R$$



【Theorem 2】 Let *R* be an equivalence relation on a set *A*. Then the equivalence classes of *R* form a partition of *A*. Conversely, given a partition $\{A_i \mid i \in I\}$ of the set *A*, there is an equivalence relation *R* that has the sets $A_i, i \in I$, as its equivalence classes.

Proof:

an equivalence relation on a set A $\leftrightarrow$ a partition of A

- **the equivalence class of *R* :** $A_1, A_2, \cdots, A_n$

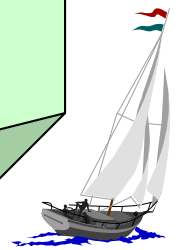
(1) $A_i \neq \phi$ for $i \in Z$

(2) $A_i \cap A_j = \phi$, when $i \neq j$

$$A_i \cap A_j \neq \phi \Rightarrow A_i = A_j$$

(3) $\bigcup_{i \in Z} A_i = A$

$$\forall a \in A, (a,a) \in R \Rightarrow \exists i, a \in A_i$$



■ $pr(A) = \{A_1, A_2, \cdots, A_n\}$

$R : \forall a, b \in A, aRb$ if and only if $a, b \in A_i$

(1) $pr(A) = \{A_1, A_2, \cdots, A_n\} \Rightarrow A = \bigcup_{i=1}^{n} A_i$

$\Rightarrow \forall a \in A, \exists i$ such that $a \in A_i \ (i = 1, 2, \cdots)$ $\therefore aRa$

(2) $aRb \Rightarrow a, b \in A_i \Rightarrow b, a \in A_i$ $\therefore bRa$

(3) aRb, bRc

$\Rightarrow \exists i, j$ such that $a, b \in A_i, b, c \in A_j$

If $i \neq j$, Then $A_i \cap A_j = \phi$

$b \in A_i$; $b \in A_j$ $\therefore A_i \cap A_j \neq \phi$

$i = j$, $a, b, c \in A_i \Rightarrow aRc$

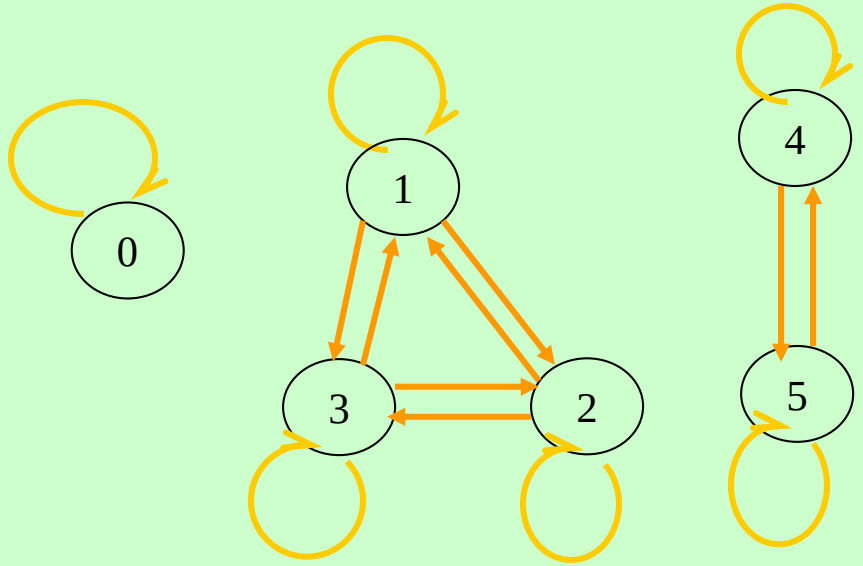


〖Example 3〗 Find the partition of the set *A* from *R*.

$A = \{0,1,2,3,4,5\}$,

$R = \{(0,0),(1,1),(2,2),(3,3),(1,2),(1,3),(2,1),(2,3),(3,1),(3,2),(4,4),(4,5),(5,4),(5,5)\}$

Solution:



[0]={0}

[1]={1,2,3}

[4]={4,5}

***pr*(*A*)={ [0], [1], [4]}**

Question:

Congruence Modulo *m*

$$R = \{(a,b) \mid a \equiv b (\bmod m), a, b \in Z\},\ pr(Z) = ?$$

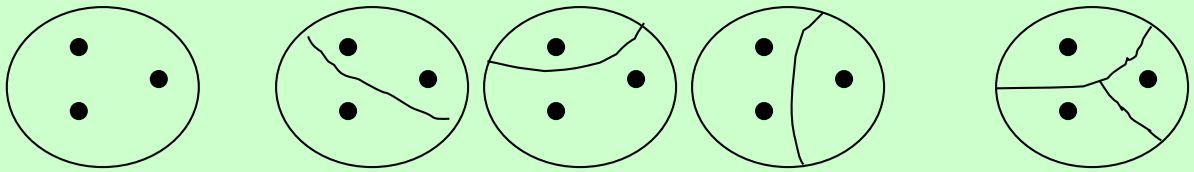
$$pr(Z) = \{[0]_m, [1]_m, \cdots, [m-1]_m\}$$

Question:

$|A|$=3. How many different equivalence relations on the set *A* are there?

Solution:

an equivalence relation on a set A $\leftrightarrow$ a partition of A



3. The operations of equivalence relations

【Theorem 3】 If R_1, R_2 are equivalence relations on A, then $R_1 \cap R_2$ is equivalence relations on A.

Proof:

$\forall a \in A \quad \because (a,a) \in R_1, (a,a) \in R_2$

$\therefore (a,a) \in R_1 \cap R_2$

It suffices to sh

- reflexive relations is reflexive,
- symmetric relations is symmetric,

and

- transitive relations is transitive.



3. The operations of equivalence relations

【Theorem 3】 If R_1, R_2 are equivalence relations on *A*, then $R_1 \cap R_2$ is equivalence relations on *A*.

If $(a,b) \in R_1 \cap R_2$

Then $(a,b) \in R_1$ and $(a,b) \in R_2$

Then $(b,a) \in R_1$ and $(b,a) \in R_2$

$(b,a) \in R_1 \cap R_2$

Proof:

It suffices to show that the inter

- **reflexive relations is reflexiv**
- **symmetric relations is symmetric,**

and

- **transitive relations is transitive.**



3. The operations of equivalence relations

【Theorem 3】 If R_1, R_2 are equivalence relations on *A*, then $R_1 \cap R_2$ is equivalence relations on *A*.

Proof:

It suffices to show th

- reflexive relation
- symmetric relatio

and

- transitive relations is transitive.

If $(a,b),(b,c) \in R_1 \cap R_2$
Then $(a,b),(b,c) \in R_1 (R_2)$
Then $(a,c) \in R_1 (R_2)$
$(a,c) \in R_1 \cap R_2$



【Theorem 4】 If R_1, R_2 are equivalence relations on *A*, then $R_1 \cup R_2$ is reflexive and symmetric relation on A.

Proof:

(1) reflexive

$$\forall a \in A \ \because (a,a) \in R_1, (a,a) \in R_2 \ \therefore (a,a) \in R_1 \cup R_2$$

(2) symmetric

$$(a,b) \in R_1 \cup R_2 \Rightarrow (a,b) \in R_1 \text{ or } (a,b) \in R_2$$
$$\Rightarrow (b,a) \in R_1 \text{ or } (b,a) \in R_2 \Rightarrow (b,a) \in R_1 \cup R_2$$

Question: transitive?



〖Example 4〗 $A=\{a,b,c\}$,

$R_1=\{(a,a),(b,b),(c,c),(a,b),(b,a)\}$

$R_2=\{(a,a),(b,b),(c,c),(b,c),(c,b)\}$

Is $R_1 \cup R_2$ a transitive relation ?

Solution:

$$R_1 \cup R_2=\{(a,a),(b,b),(c,c),(a,b),(b,a),(b,c),(c,b)\}$$



Homework:

P. 646: 3, 8, 16, 24, 40, 68



Chapter 9 Relations

9.1 Relations and Their Properties

9.2 n-ary Relations and Their Application

9.3 Representing Relations

9.4 Closures of Relations

9.5 Equivalence Relations

9.6 Partial Orderings



1. Partial Orderings

【Definition】 Let R be a relation on S. Then R is a *partial ordering or partial order* if R is

- reflexive
- antisymmetric
- transitive

Notation: (S, R) ---- partially ordered set or a *poset*

〖Example 1〗 (1) $R_1 = \{(a,b) \mid a \le b, a, b \in Z\}$ $(Z, \le)$

(2) $R_2 = \{(a,b) \mid a \mid b, a, b \in Z^+\}$ $(Z^+, |)$

(3) $R_3 = \{(s_1, s_2) \mid s_1 \subseteq s_2, s_1, s_2 \in P(S)\}$ $(P(S), \subseteq)$



Notation:

$$a \prec = b ----(S, R) \quad is \quad a \quad poset, \quad (a,b) \in R$$

【Definition】*comparable/ incomparable*

The elements *a* and *b* of a poset $(S, \leq)$ are called *comparable* if either $a \leq b$ or $b \leq a$. When *a* and *b* are elements of *S* such that neither $a \leq b$ nor $b \leq a$, *a* and *b* are called *incomparable*.

For example,

$$(Z^+, |) \qquad 2 \nmid 7 \qquad 2 \mid 4$$



【Definition】If $(S,\leq)$ is a poset and every two elements of *S* are comparable, *S* is called a *totally ordered* or *linearly ordered set*, $\leq$ is called a *total order* or *linear order*. In this case $(S,\leq)$ is called a *chain*.

〖Example 2〗

(1) $(Z,\leq)$ is a poset. In this case either $a \leq b$ or $b \leq a$.
Hence, $\leq$ is a total order and $(Z,\leq)$ is a chain.

(2) $(Z^+,|)$ is a poset, not a totally ordered set.

(3) $(P(S),\subseteq)$ is a poset, not a totally ordered set.



2. Lexicographic Order

The lexicographic order is a special case of an ordering of strings on a set constructed from a partial ordering on the set.

Given two posets $(A_1, \leq_1)$ **and** $(A_2, \leq_2)$**, we construct an *induced* partial order *R* on** $A_1 \times A_2$**:** $(x_1, y_1) < (x_2, y_2)$ **if**

$$x_1 <_1 x_2$$

or

$$x_1 = x_2 \quad \textit{and} \quad y_1 <_2 y_2$$



A *lexicographic ordering* is a partial ordering defined on a Cartesian product of two posets.

Proof: $(A_1 \times A_2, \leq)$

① **reflexive**

$$\forall (a,b) \in A_1 \times A_2, \qquad (a,b) \leq (a,b)$$

② **antisymmetric**

$$\left.\begin{array}{l}(a_1,b_1) \leq (a_2,b_2)\\ (a_1,b_1) \neq (a_2,b_2)\end{array}\right\} \Rightarrow a_1 \leq_1 a_2 \qquad or \qquad a_1 = a_2, b_1 \leq_2 b_2$$

③ **transitive**

$$\left.\begin{array}{l}(a_1,b_1) \leq (a_2,b_2)\\ (a_2,b_2) \leq (a_3,b_3)\end{array}\right\} \Rightarrow (a_1,b_1) \leq (a_3,b_3)$$



〖Example 3〗 Let $A_1 = A_2 = Z^+$ *and* $R_1 = R_2 = |$ **. Then**

(1) (2, 4) < (2, 8) since $x_1 = x_2$ *and* $y_1 R_2 y_2$

(2) (2, 4) is not related under *R* to (2, 6) since $x_1 = x_2$ **but 4 does not divide 6.**

(3) (2, 4) < (4, 5) since $x_1 R_1 x_2$

This definition of lexicographic order extends naturally to multiple Cartesian products of partially ordered sets.

〖Example 4〗 Using the same definitions of A_i *and* R_i **as above,**

(1) (2, 3, 4, 5) < (2, 3, 8, 2) since $x_1 = x_2$ *and* $y_1 = y_2$ **and 4 divides 8.**

(2) (2, 3, 4, 5) is not related to (3, 6, 8, 10) since 2 does not divide 3.



lexicographic order of string

The string $a_1 a_2 \cdots a_m$ is less then $b_1 b_2 \cdots b_n$ if and only if

$$(a_1, a_2, \cdots, a_t) < (b_1, b_2, \cdots, b_t), or$$

$$(a_1, a_2, \cdots, a_t) = (b_1, b_2, \cdots, b_t) \quad and \quad m < n$$

Where $t = \min(m, n)$

For example,

Find the lexicographic ordering of the following strings.

discrete discredit discreet discreteness discretion

discredit < discreet < discrete < discreteness < discretion



3. Hasse Diagrams

Hasse Diagrams

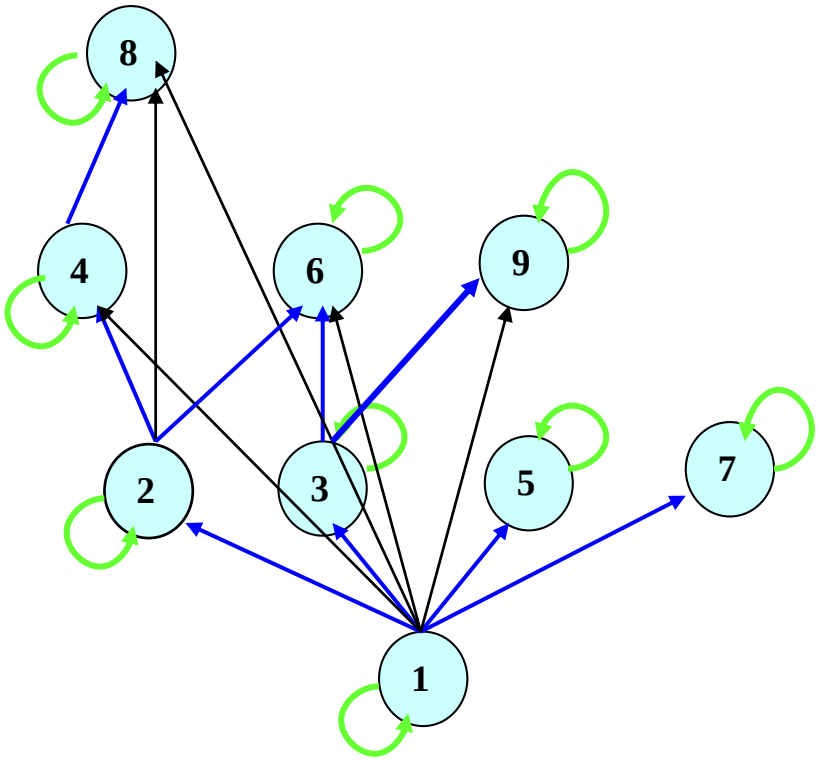
---- A method used to represent a partial ordering

To construct a Hasse diagram:

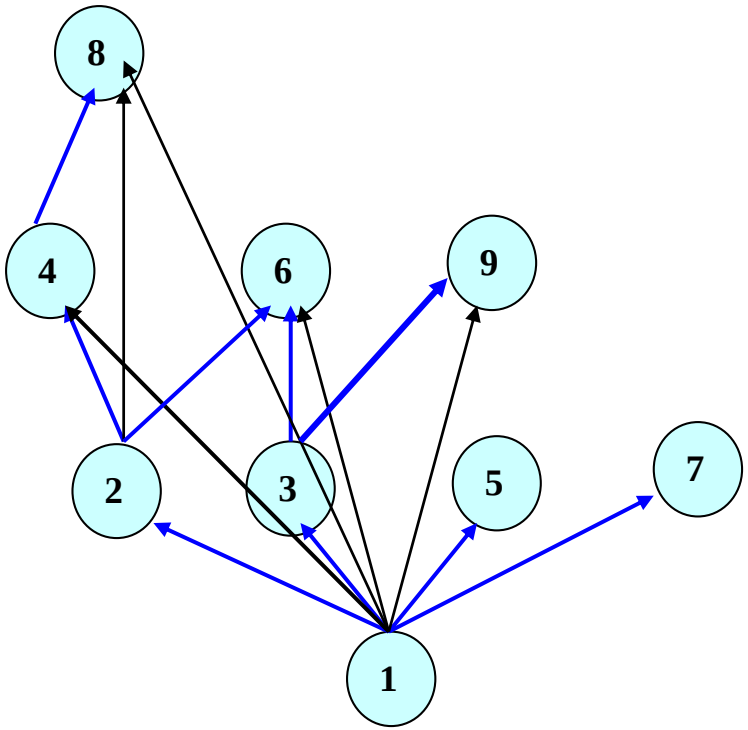
1) **Construct a digraph representation of the poset *(A, R)* so that all arcs point up (except the loops).**
2) **Eliminate all loops**
3) **Eliminate all arcs that are redundant because of transitivity**
4) **Eliminate the arrows at the ends of arcs since everything points up.**



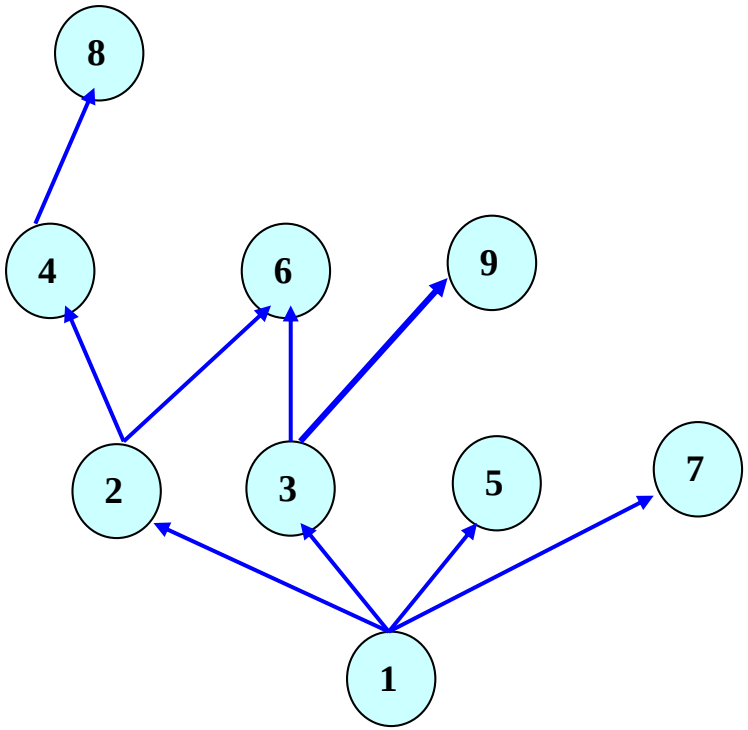
〖**Example 5**〗 $A = \{1,2,3,4,5,6,7,8,9\}$ $R = \{(a,b) \mid a \mid b, a, b \in A\}$



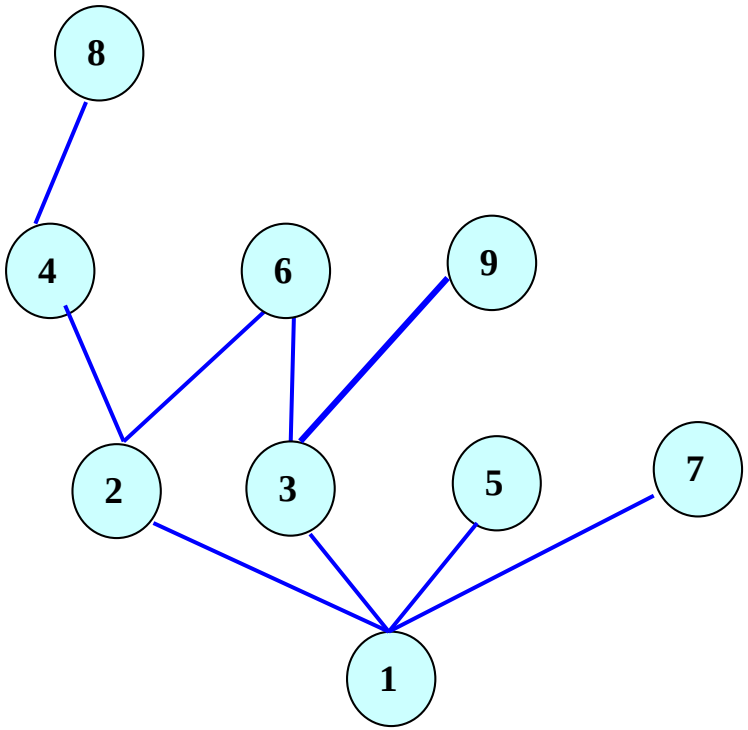
〖**Example 5**〗 $A = \{1,2,3,4,5,6,7,8,9\}$ $R = \{(a,b) \mid a \mid b, a, b \in A\}$



〖Example 5〗 $A = \{1,2,3,4,5,6,7,8,9\}$ $R = \{(a,b) \mid a \mid b, a, b \in A\}$



〖**Example 5**〗 $A = \{1,2,3,4,5,6,7,8,9\}$ $R = \{(a,b) \mid a \mid b, a, b \in A\}$



〖Example 6〗

$A = \{1,2,3,4,5,6,7,8,9\}$

(1) $R_1 = \{(a,b) \mid a \le b, a, b \in A\}$

(2) $R_2 = \{(a,b) \mid a \mid b, a, b \in A\}$

(3) $A = \{a,b,c\}, R_3 = \{(s_1, s_2) \mid s_1 \subseteq s_2, s_1, s_2 \in P(A)\}$



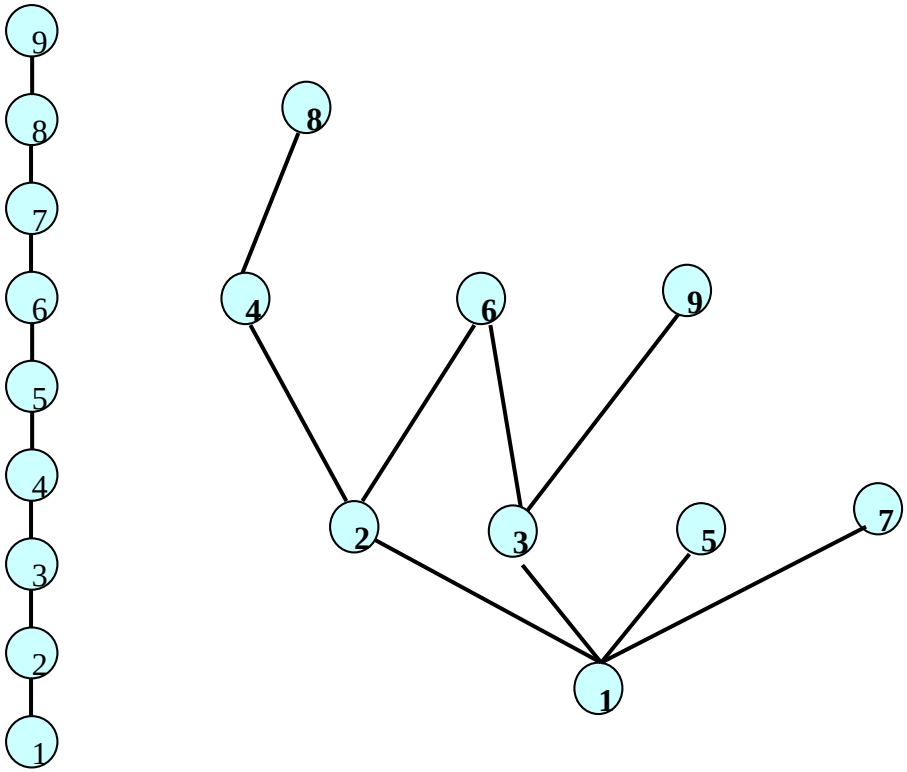
〖Example 6〗

$A = \{1,2,3,4,5,6,7,8,9\}$

(1) $R_1 = \{(a,b) \mid a \le b, a, b \in A\}$

(2) $R_2 = \{(a,b) \mid a \mid b, a, b \in A\}$

(3) $A = \{a,b,c\}, R_3 = \{(s_1, s_2) \mid s_1 \subseteq s_2, s_1, s_2 \in P(A)\}$



〖Example 6〗

$A = \{1,2,3,4,5,6,7,8,9\}$

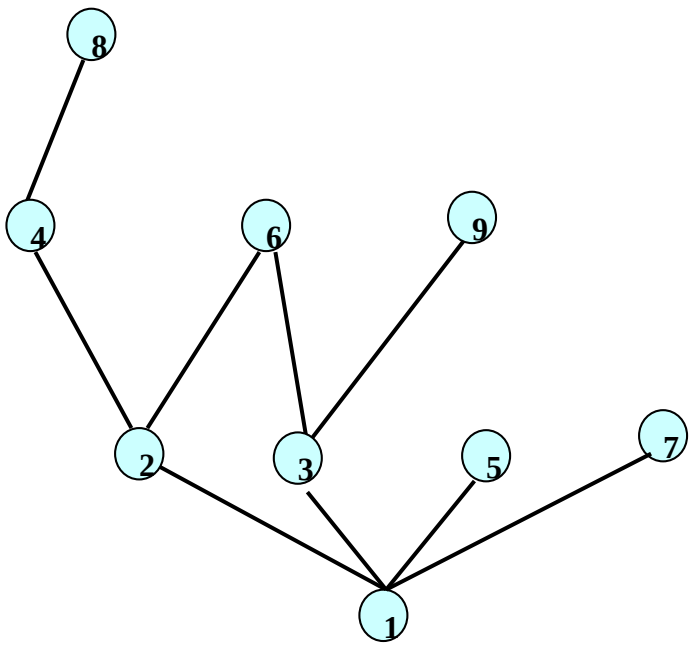
(1) $R_1 = \{(a,b) \mid a \le b, a,b \in A\}$

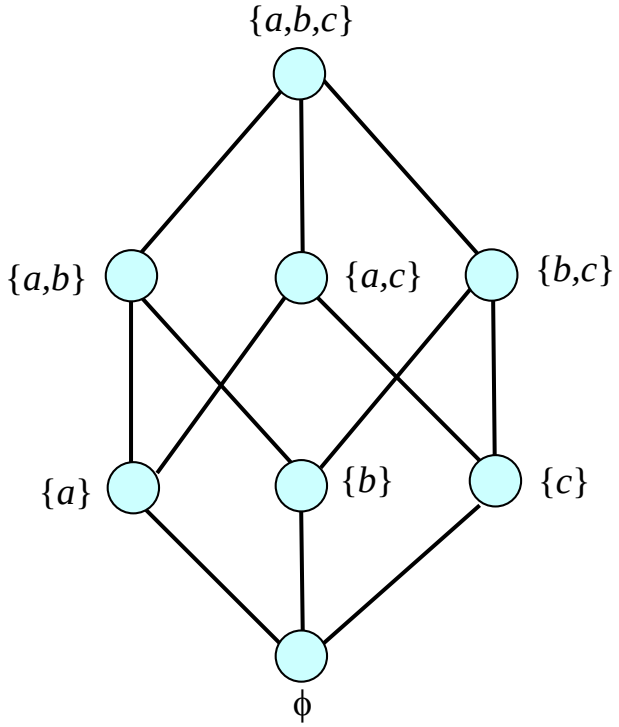
(2) $R_2 = \{(a,b) \mid a \mid b, a,b \in A\}$

(3) $A = \{a,b,c\}, R_3 = \{(s_1,s_2) \mid s_1 \subseteq s_2, s_1,s_2 \in P(A)\}$

$p(A) = \{\phi,\{a\},\{b\},\{c\},\{a,b\},\{b,c\},\{a,c\},\{a,b,c\}\}$







4. chain and antichain

【Definition】 $(A,\leq)$ **is a poset.** $B \subseteq A$ **，if** $(B,\leq)$ **is a totally ordered set，the *B* is called a *chain* of** $(A,\leq)$ **.**

The length of chain： $|B|$**, *B* is a definite set，**

$B \subseteq A$ **，if** $\forall a,b \in B(a \neq b),(a,b) \notin R,(b,a) \notin R$

the *B* is called a *antichain* of $(A,\leq)$ **.**

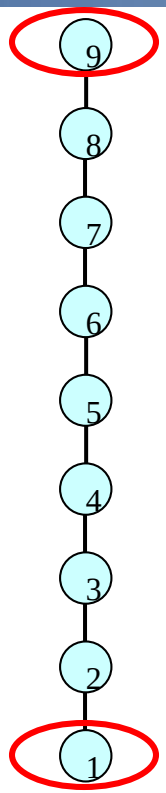


5. Maximal and Minimal Elements

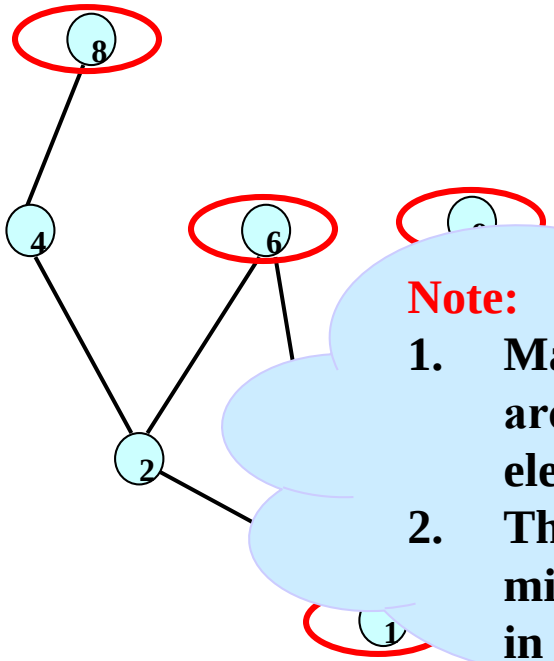
【Definition】Let $(A, \leq)$ be a poset. $a \in A$, then *a* is a *maximal element* if there does not exist an element *b* in *A* such that $a \prec b$.
Similarly for a *minimal element*.

For example,

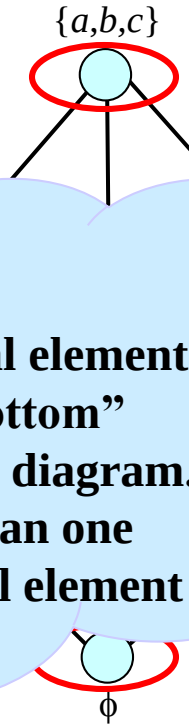




(1)



(2)



(3)

Note:

1. **Maximal and minimal elements are the "top" and "bottom" elements in the Hasse diagram.**
2. **There can be more than one minimal and maximal element in a poset.**

	maximal element	minimal element
(1)	9	1
(2)	8,6,9,5,7	1
(3)	$\{a, b, c\}$	ϕ

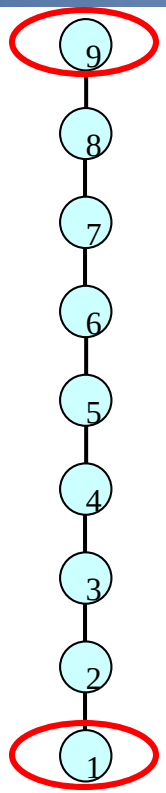


6. Greatest and Least Element

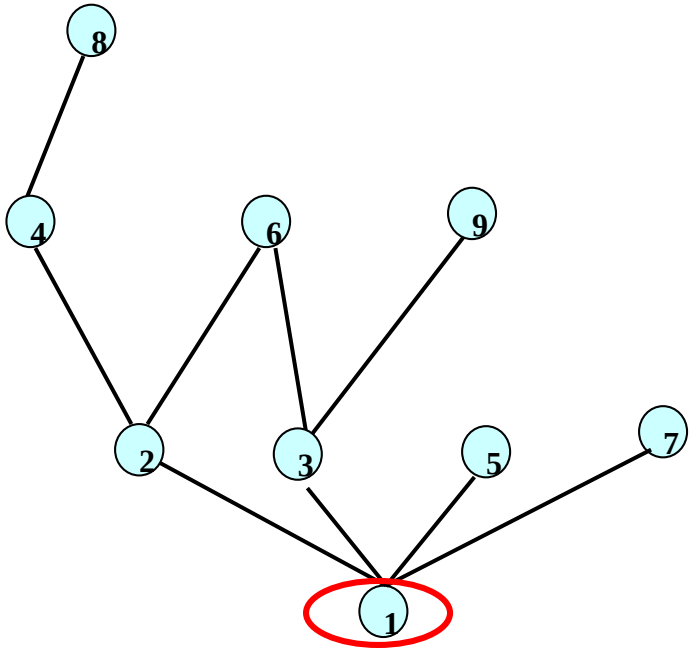
【Definition】Let $(A, \leq)$ be a poset. Then an element *a* in *A* is a *greatest element* of *A* if $b \leq a$ for every *b* in *A*, and *a* is a *least element* of *A* if $a \leq b$ for every *b* in *A*.

For example,

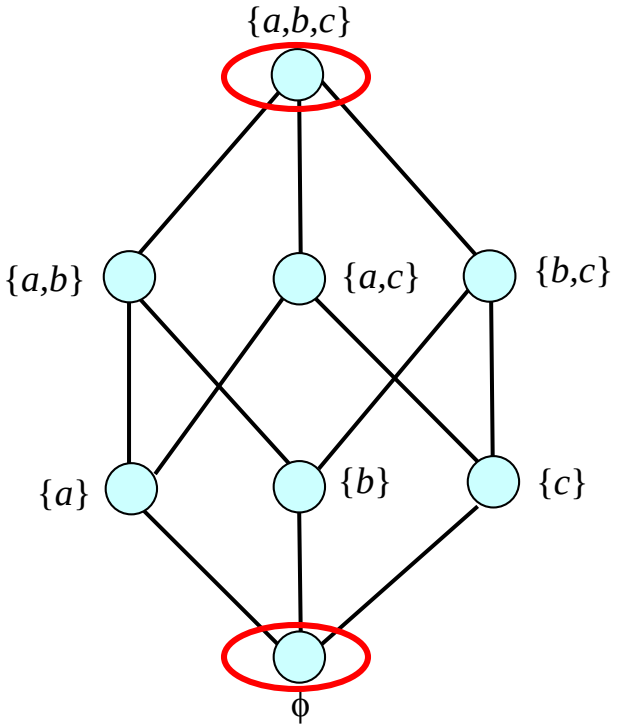




(1)



(2)



(3)

	The greatest element	The least element
(1)	**9**	**1**
(2)	**/**	**1**
(3)	**{*a*, *b*, *c*}**	**ϕ**

【 Theorem 】 The greatest and least element are unique when they exist.

Proof:

Suppose that a_1 is a greatest element in A. It follows that

$x \le a_1$ **for every x in A.**

Suppose that a_2 is a greatest element in A. It follows that

$x \le a_2$ **for every x in A.**

It implies that $a_2 \le a_1 \; and \; a_1 \le a_2$

That is $a_1 = a_2$

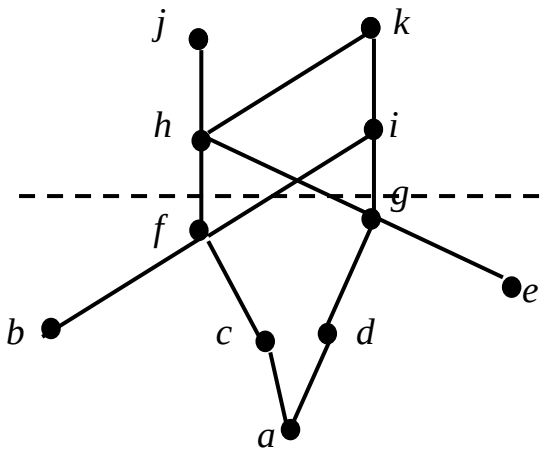


7. Upper and Lower Bounds

【Definition】Let *A* be a subset of *S* in the poset $(S, \leq)$. If there exists an element *a* in *S* such that $b \leq a$ for all *b* in *A*, then *a* is called an *upper bound* of *A*.
Similarly for *lower bounds*.

〖Example 7〗 $S = \{a, b, c, d, e, f, g, h, i, j, k\}$

$A = \{a, b, c, d, e, f, g\}, A' = \{h, i, j, k\}$



the upper bounds of set *A*: *h, i, k, j*

the lower bounds of set A: /

the upper bounds of set A': /

the lower bounds of set A': *f, g, a, b, c, d, e*

8. Least Upper and Greatest Lower Bounds

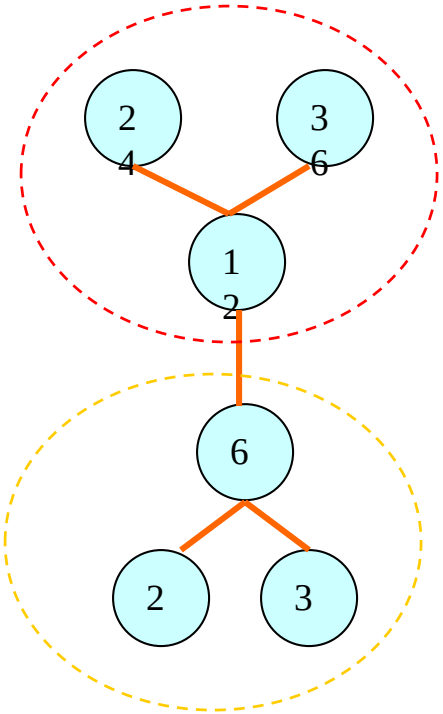
【Definition】If *a* is an upper bound for *P* which is less than every other upper bounds then it is the *least upper bound*, denoted by lub(*S*).

Similarly for the *greatest lower bound*, glb(*S*).



〖Example 7〗 $A=\{2,3,6,12,24,36\}$, $B_1=\{2,3,6\}$, $B_2=\{12,24,36\}$, R :|

Determine the maximal elements, minimal elements, greatest element, least element of set *A*, the upper bounds, lower bounds , least upper bound , greatest lower bound of set B_1, B_2 。



***A*:**

maximal elements: 24,36 **minimal elements:2,3**

the greatest element: / **the least element: /**

***B*$_1$:**

upper bounds: 6,12,24,36 **lower bounds: /**

The least upper bound: 6 **The greatest lower bound:/**

***B*$_2$:**

uper bounds: / **lower bounds: 12,6,2,3**

The least upper bound: / **The greatest lower bound:12**

9. Well-ordered Sets

【Definition】**A poset *(A, R)* is *well-ordered set* if every nonempty subset of *A* has a least element.**

For example,

$(1) A = \{1, 2, \cdots, n\}, (A, \leq)$

$(2) A = \{x \mid 0 < x < 1\}, (A, \leq)$

Note:

A well-ordered set is a totally ordered set.

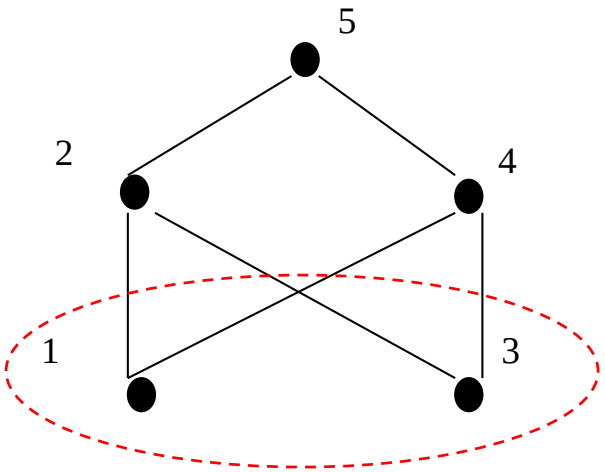


10. Lattices

【Definition】A poset is called a *lattice* if every pair of elements has a lub and a glb.

〖Example 9〗 Determine whether the following poset is lattice.

(1)



(2) $(Z, \leq)$ **?**

lub: the larger of the two elements,

glb: the smaller of the two elements.

Hence, the poset $(Z, \leq)$ **is a lattice.**

Note: Every totally ordered set is a lattice.

(3) $(Z^+, |)$ **?**

$\forall a, b \in Z^+$

lub: the least common multiple,

glb: the greatest common divisor

Hence, the poset $(Z^+, |)$ **is a lattice.**

(4) $(P(s), \subseteq)$ **?**

$\forall s_1, s_2 \in P(s)$ **, lub:** $s_1 \bigcup s_2$ **glb:** $s_1 \bigcap s_2$

Hence, the poset $(P(s), \subseteq)$ **is a lattice.**



Page 659, Example 25: The lattice model of information flow

The Lattice Model of Information Flow In many settings the flow of information from one person or computer program to another is restricted via security clearances. We can use a lattice model to represent different information flow policies. For example, one common information flow policy is the *multilevel security policy* used in government and military systems. Each piece of information is assigned to a security class, and each security class is represented by a pair (A, C) where A is an *authority level* and C is a *category*. People and computer programs are then allowed access to information from a specific restricted set of security classes.

The typical authority levels used in the U.S. government are unclassified (0), confidential (1), secret (2), and top secret (3). (Information is said to be classified if it is confidential, secret, or top secret.) Categories used in security classes are the subsets of a set of all *compartments* relevant to a particular area of interest. Each compartment represents a particular subject area. For example, if the set of compartments is {*spies, moles, double agents*}, then there are eight different categories, one for each of the eight subsets of the set of compartments, such as {*spies, moles*}.

We can order security classes by specifying that $(A_1, C_1) \preccurlyeq (A_2, C_2)$ if and only if $A_1 \leq A_2$ and $C_1 \subseteq C_2$. Information is permitted to flow from security class (A_1, C_1) into security class (A_2, C_2) if and only if $(A_1, C_1) \preccurlyeq (A_2, C_2)$. For example, information is permitted to flow from the security class (*secret,* {*spies, moles*}) into the security class (*top secret,* {*spies, moles, double agents*}), whereas information is not allowed to flow from the security class (*top secret,* {*spies, moles*}) into either of the security classes (*secret,* {*spies, moles, double agents*}) or (*top secret,* {*spies*}).

We leave it to the reader (see Exercise 48) to show that the set of all security classes with the ordering defined in this example forms a lattice.



11. Topological Sorting

The problem of project schedule:

Suppose that a project is made up of *n* different tasks. Some tasks can be completed only after others have been finished. How can an order be found for these tasks?

We impose a total ordering $\leq$ on a poset *compatible* with the partial order if $a \leq b$ whenever aRb .

(*A*,*R*) is a poset. (*A*,$\leq$) is a total ordering.

If $a \leq b$ whenever aRb .

The total ordering $\leq$ is called compatible with the partial ordering R.



Constructing a compatible total ordering from a partial ordering is called *topological sorting*.

【Lemma 1】 Every finite nonempty poset $(S, \leq)$ has a minimal element.

Proof:

Choose an element a_0 of S.

If a_0 is not minimal, then there is an element a_1 with $a_1 < a_0$.

If a_1 is not minimal, there is an element a_2 with $a_2 < a_1$.

Continue this process, so that

If a_n is not minimal, there is an element a_{n+1} with $a_{n+1} < a_n$.

Since there are only finite number of elements in the poset, this process must end with a minimal element.

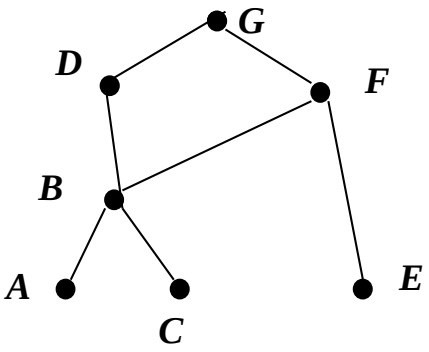


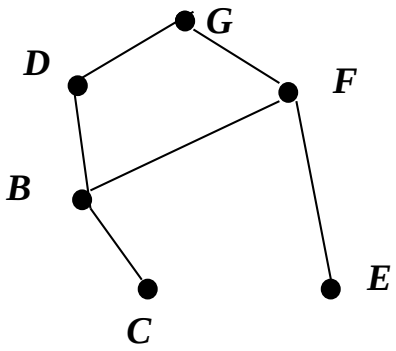
Algorithm: To sort a poset (S, R).

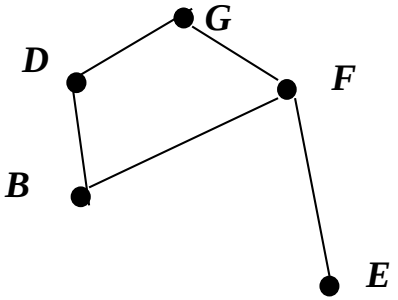
- **Select a (any) minimal element and put it in the list. Delete it from S.**
- **Continue until all elements appear in the list (and S is void).**

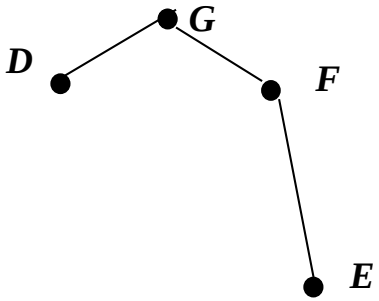


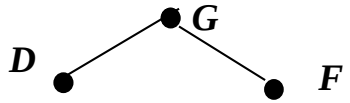
〖Example 27〗A project is broken down into seven tasks. A partial ordering on tasks is set up by considering task *x* < task *y* if task *y* can't be started until task *x* has been completed.
Find an ordered in which these tasks can be carried out to complete the project.



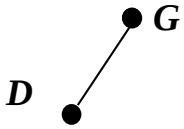








The order for the task:
A<C<B<E<F<D<G





Homework:

P. 662: 5, 12, 24, 32, 44, 54

